

Project Initialization and Planning Phase

Date	02 July 2026
Team ID	xxxxxxx
Project Title	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed Credit Card Approval Prediction System aims to automate the process of approving or rejecting credit card applications using Machine Learning techniques. The system improves efficiency and accuracy by analyzing applicant details and predicting approval status in real-time. This helps financial institutions reduce manual effort, minimize errors, and make faster decisions.

Project Overview	
Objective	The primary objective of this project is to develop an intelligent system that predicts whether a credit card application will be approved or rejected based on customer information and credit history using Machine Learning algorithms.
Scope	The scope of this project includes data collection, preprocessing, model training, and deployment of a web-based application using Flask to provide real-time prediction for credit card approvals.
Problem Statement	
Description	Traditional credit card approval systems involve manual verification which is time-consuming, error-prone, and inefficient. This project addresses these limitations by introducing an automated machine learning-based prediction system.
Impact	Implementing this solution will improve approval speed, reduce operational workload, enhance decision accuracy, and increase customer satisfaction in financial institutions.
Proposed Solution	
Approach	The system uses Machine Learning models to analyze applicant data such as income, education, family status, employment details, and credit records to predict credit card approval status. The trained model is integrated into a Flask web application for easy user interaction.

Key Features	- Machine learning-based approval prediction - Real-time user input and prediction - Fast and accurate decision making - Web-based interface using Flask - Reduced manual verification process
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	- Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Application Record Dataset (application_record.csv) Credit Record Dataset (credit_record.csv)